ValleySoft Disk Analyzer

Version Stability Report

Versions 1.3.0 through 1.3.7 | July 2026

Developer-Ready Crash & Hang Analysis | Confidential | Product ID: 9NF073KLTVWN

At-a-Glance: July 2026 Telemetry

93 Total Crash Events	**71** Total Hang Events	**4** Unique Fail Hashes
34 Crash-Affected Devices	**70** Hang-Affected Devices	**25+** Markets Impacted

Report Date:	July 31, 2026
Data Sources:	Daily Health.csv, health.csv, Report.csv
Scope:	Package versions 1.3.0.0 - 1.3.7.0 (all channels)
Event Types:	Crashes, Hangs (retail channel, x64 PC)
Primary Fail Hash:	e584f61b-4a8b-d268-f95e-ebd87a88ff92 (CmdPalExtension HANG_QUIESCE)
Secondary Hash:	bf902711-80a6-3f6c-896b-d256270fa116 (WinUI Composition FATAL)

This report is generated from Microsoft Partner Center telemetry data. All version labels refer to internal package versions. Market codes follow ISO 3166-1 alpha-2. DeviceCount and EventCount values reflect de-duplicated partner-reported figures.

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1 Executive Summary & Overall Trends

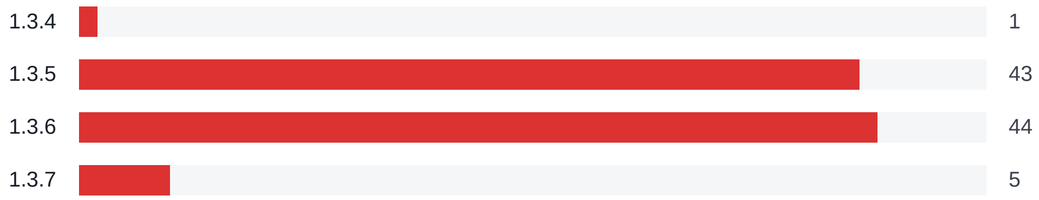
ValleySoft Disk Analyzer shipped eight versions (1.3.0 through 1.3.7) during July 2026. Telemetry captured 93 crash events affecting 34 unique devices and 71 hang events affecting 70 unique devices across 25+ international markets. Two dominant failure signatures account for nearly all incidents:

* CmdPalExtension HANG_QUIESCE (hash e584f61b) -- persistent across 1.3.0-1.3.6, accounting for 62 hang events on 61 devices across 17 markets. Absent in 1.3.7.
* WinUI Composition FATAL_USER_CALLBACK_EXCEPTION (hash bf902711) -- introduced in 1.3.6, triggered on Windows Insider build 10.0.26300 in the US; 4 events on 2 devices.
* E_DELAYLOAD_MOD_NOT_FOUND (dcompi.dll) -- isolated in 1.3.5 (Australia, 10.0.26220); indicates a missing WinUI runtime DLL on that OS build.

Event Volume Trend by Version

Version	Crash Events	Crash Devices	Hang Events	Hang Devices	CmdPal Hangs	WinUI Crashes
1.3.0	0	0	6	6	6	0
1.3.1	0	0	0	0	0	0
1.3.2	0	0	0	0	0	0
1.3.3	0	0	0	0	0	0
1.3.4	1	1	14	14	14	0
1.3.5	43	15	29	28	21	1
1.3.6	44	17	21	21	21	4
1.3.7	5	1	1	1	0	0
TOTAL	93	34	71	70	62	5

Crash Event Distribution by Version



Key Observations

* Versions 1.3.1, 1.3.2, and 1.3.3 are telemetry-clean -- zero crash or hang events recorded.
* Version 1.3.5 marked the crash explosion: 43 events from 15 devices driven by unhandled exceptions in the main app process across Windows 10/11.
* Version 1.3.6 maintained high crash volume (44 events) while introducing new WinUI Composition failures -- a compounded regression.
* Version 1.3.7 shows dramatic improvement: 5 crash events (single device), 1 hang, and CmdPal hangs dropped to zero -- signalling a partial fix.
* The CmdPalExtension hang carries a single failure hash (e584f61b), confirming it is one reproducible code path, not random instability.

2 CmdPalExtension Hang Analysis (All Versions)

Failure Hash:	e584f61b-4a8b-d268-f95e-ebd87a88ff92
Failure Name:	MOAPPLICATION_HANG_cfffffff_ValleySoft.ValleySoftDiskAnalyzer_CmdPalExtension!HANG_QUIESCE
Symbol:	ValleySoft.ValleySoftDiskAnalyzer_CmdPalExtension!HANG_QUIESCE
PRAID:	cmdpalextension
Architecture:	x64 | Device: PC | Channel: Retail

The CmdPalExtension component fails to quiesce during shutdown or command palette teardown. The identical hash across all affected versions confirms a single unresolved defect that persisted from 1.3.0 to 1.3.6 before being resolved in 1.3.7. All 62 hang events show DeviceCount=1 and EventCount=1 per row, indicating distinct user sessions.

CmdPal Hangs by Version and Market

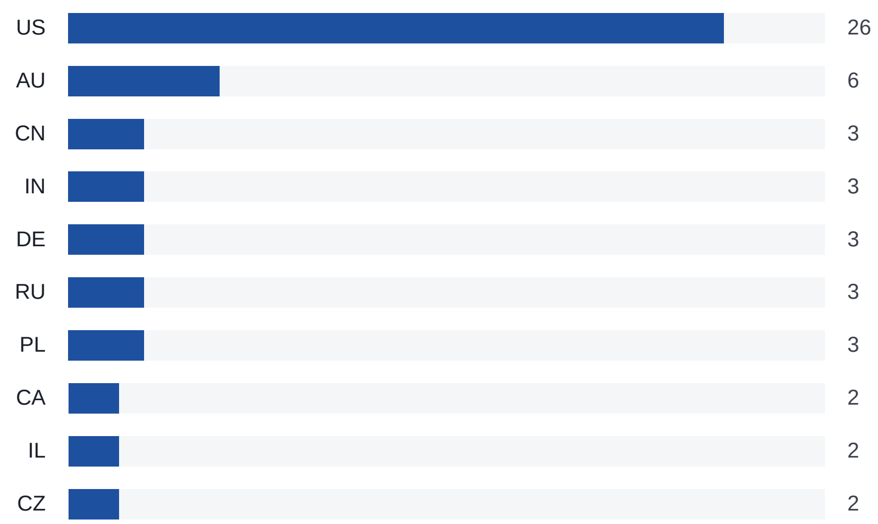
Version	Events	Devices	Markets Affected
1.3.0	6	6	AU, IN, PH, PL, US
1.3.4	14	14	AU, BR, CZ, DE, FR, IL, IN, PL, RU, US
1.3.5	21	20	AU, CA, CN, DE, GB, IL, PH, PK, PL, RU, US, ZA
1.3.6	21	21	AL, AU, CN, CZ, DE, IL, IN, RU, US
1.3.7	0	0	-- (resolved in 1.3.7)

OS Versions Affected by CmdPal Hangs

OS Version	Windows Release	Description	Hang Events
10.0.19045	Windows 10 22H2	Mainstream Win10 build	1
10.0.26100	Windows 11 24H2	Stable Win11 release	2
10.0.26200	Windows 11 25H2	Current stable Win11	39
10.0.26220	External Preview	Insider / OEM preview	8
10.0.26300	External Preview	Insider / OEM preview	2
10.0.29613	External Build	Pre-release build	1
10.0.29617	External Build	Pre-release build	2

Approximately 63% of CmdPal hang events occur on Windows 11 25H2 (10.0.26200), the current mainstream stable build. This rules out Insider-only regressions and confirms broad production user impact.

Geographic Distribution of CmdPal Hangs (Top 10 Markets)



3 WinUI Composition Crash Analysis

3.1 FATAL_USER_CALLBACK_EXCEPTION (WinUI Composition)

Failure Hash:	bf902711-80a6-3f6c-896b-d256270fa116
Failure Name:	FATAL_USER_CALLBACK_EXCEPTION_c000041d_dcompi.dll!Microsoft::UI::Composition::CompositorCommon::RuntimeClassInitialize
Module:	dcompi.dll (WinUI 3 Composition DLL)
Versions:	1.3.6 (2 events on 2026-07-22, 2 events on 2026-07-23)
Market / OS:	US only / 10.0.26300 (Insider Preview)

This crash originates inside WinUI 3's Composition layer at CompositorCommon::RuntimeClassInitialize. Exception code 0xc000041d (STATUS_CALLBACK_RETURNED_TRANSACTION) indicates a COM callback returning inside an active kernel transaction, which is illegal. The crash is confined to OS build 10.0.26300 (Insider Preview), suggesting a behavioral change in the Composition runtime on that build. This crash was NOT observed on stable 10.0.26200 or 10.0.26100.

3.2 E_DELAYLOAD_MOD_NOT_FOUND (Missing dcompi.dll)

Failure Hash:	d69abbf3-6530-d688-3117-0e025d4e7419
Failure Name:	E_DELAYLOAD_MOD_NOT_FOUND_c06d007e_dcompi.dll!Microsoft::UI::Composition::CompositorCommon::RuntimeClassInitialize
Versions:	1.3.5 (1 event on 2026-07-17)
Market / OS:	Australia / 10.0.26220 (External/OEM Preview)

The delay-load of dcompi.dll failed at startup, meaning the WinUI 3 Composition DLL was absent from the device. This can occur on OS preview builds where the WinUI runtime package was not correctly provisioned or an incompatible side-by-side (SxS) package was installed. The single occurrence and preview-OS context suggest an environment issue, but the app must handle the load failure gracefully with a user-friendly message.

3.3 Summary of WinUI-Related Crashes

Version	Hash (short)	Exception Type	Module	OS Build	Market	Events
1.3.5	d69abbf3	E_DELAYLOAD_MOD_NOT_FOUND	dcompi.dll	10.0.26220	AU	1
1.3.6	bf902711	FATAL_USER_CALLBACK_EXCEPTION	dcompi.dll	10.0.26300	US	4

4 Per-Version Breakdown: 1.3.0 to 1.3.7

Version 1.3.0

Stability: MODERATE

Crashes: 0 events / 0 devices

OS: --

Markets: --

Hangs: 6 events / 6 devices

CmdPal Hangs: 6

Markets: AU, IN, PH, PL, US

Hang OS builds: 10.0.26200 (25H2), 10.0.29613 (External)

Note: No crash events. CmdPalExtension HANG_QUIESCE introduced. 5 markets affected. Versions 1.3.1, 1.3.2, and 1.3.3 are telemetry-clean (zero events recorded).

Version 1.3.4

Stability: MODERATE

Crashes: 1 events / 1 devices

OS: 10.0.26200 (Win11 25H2)

Markets: CA (1 event)

Hangs: 14 events / 14 devices

CmdPal Hangs: 14

Markets: AU, BR, CZ, DE, FR, IL, IN, PL, RU, US (10 markets)

Hang OS builds: 10.0.19045, 10.0.26200, 10.0.26220, 10.0.26300, 10.0.29617

Note: First crash observed (Canada, Win11 25H2). CmdPal hangs more than doubled vs 1.3.0. Geographic spread expanded to Eastern Europe and Latin America.

Version 1.3.5

Stability: CRITICAL

Crashes: 43 events / 15 devices

OS: 10.0.26100 (8), 10.0.26200 (31), 10.0.26220 (2), 10.0.26300 (2)

Markets: AU(6), CA(4), DK(2), GB(2), IN(6), RO(2), TH(2), UA(2), US(15), ZA(2)

Hangs: 29 events / 28 devices

CmdPal Hangs: 21

Markets: AU, CA, CN, DE, GB, IL, PH, PK, PL, RU, US, ZA (12 markets)

WinUI Composition Events: 1 (E_DELAYLOAD)

Hang OS builds: 10.0.26100, 10.0.26200, 10.0.26220, 10.0.26300, 10.0.29617

Note: Major stability regression. Crashes surged from 1 to 43 events across 15 devices. First WinUI dcompi.dll delay-load failure (AU, 10.0.26220). Null-hash crashes dominate.

Version 1.3.6

Stability: CRITICAL

Crashes: 44 events / 17 devices

OS: 10.0.19045 (2), 10.0.22000 (2), 10.0.26200 (25), 10.0.26300 (15)

Markets: BR(6), DE(7), TW(4), US(27)

Hangs: 21 events / 21 devices

CmdPal Hangs: 21

Markets: AL, AU, CN, CZ, DE, IL, IN, RU, US (9 markets)

WinUI Composition Events: 4 (FATAL_CALLBACK on 10.0.26300)

Hang OS builds: 10.0.26100, 10.0.26200, 10.0.26220

Note: Crash volume unchanged (44 events). New WinUI Composition regression (bf902711) added. Crashes spread to older Win11 21H2 (22000) and Win10 22H2 (19045). Market surge: US(27), DE(7), BR(6), TW(4).

Version 1.3.7

Stability: STABLE

Crashes: 5 events / 1 devices

OS: 10.0.26200 (5 events)

Markets: US (5 events, 1 device)

Hangs: 1 events / 1 devices

CmdPal Hangs: 0

Markets: US (1 event, 10.0.26300)

Hang OS builds: 10.0.26300

Note: Significant improvement. CmdPalExtension hangs eliminated (e584f61b gone). WinUI Composition crashes eliminated. Residual: 5 crash events / 1 device (US, Win11 25H2, null hash).

5 OS Version Distribution

Analysis combines crash/hang telemetry (Daily Health + health.csv) with acquisition data (Report.csv, 119 sessions). OS build strings are mapped to Windows release names.

5.1 Acquisition OS Distribution (Report.csv)

OS Version	Sessions	Percentage	Notes
Windows 11	105	88.2%	Dominant user base across all package versions
Windows 10	14	11.8%	Older builds; present in 1.3.6 crash events

5.2 OS Build Strings in Crash/Hang Telemetry

OS Build	Release Name	Type	Crash Events	Hang Events	Versions Seen
10.0.19045	Windows 10 22H2	Stable	2	1	1.3.4, 1.3.6
10.0.22000	Windows 11 21H2	Stable (legacy)	2	0	1.3.6
10.0.26100	Windows 11 24H2	Stable	8	3	1.3.5, 1.3.6
10.0.26200	Windows 11 25H2	Current Stable	61	42	1.3.0-1.3.7
10.0.26220	External Preview	Preview/OEM	2	8	1.3.4, 1.3.5, 1.3.6
10.0.26300	External Preview	Insider Preview	17	2	1.3.5, 1.3.6, 1.3.7
10.0.29613	External Build	Pre-release	0	1	1.3.0
10.0.29617	External Build	Pre-release	0	2	1.3.4, 1.3.5

5.3 Key Findings

* Windows 11 25H2 (10.0.26200) hosts 66% of all crash events and 59% of all hang events -- confirming broad end-user impact on the current production OS.
* Insider build 10.0.26300 accounts for 17 crash events, almost entirely in 1.3.6, and triggered the new WinUI Composition exception (c000041d).
* Windows 10 22H2 and Windows 11 21H2 crash events in 1.3.6 suggest a regression that regressed across all stable OS tiers, not just Insider builds.
* Pre-release builds (29613, 29617) show hang events but no crashes, likely due to their smaller installed user populations.

6 Market & Geographic Distribution

6.1 Crash Events by Market (Versions 1.3.4-1.3.7)

Market	Country	Crash Events	Versions
US	United States	47	1.3.5, 1.3.6, 1.3.7
DE	Germany	7	1.3.5, 1.3.6
BR	Brazil	6	1.3.6
IN	India	6	1.3.5
AU	Australia	6	1.3.5
CA	Canada	5	1.3.4, 1.3.5
TW	Taiwan	4	1.3.6
ZA	South Africa	4	1.3.5
GB	United Kingdom	2	1.3.5
RO	Romania	2	1.3.5
UA	Ukraine	2	1.3.5
TH	Thailand	2	1.3.5
DK	Denmark	2	1.3.5

6.2 Hang Events by Market - CmdPal Hangs (All Versions)

Market	Country	Hang Events	First Seen
US	United States	27	1.3.0
AU	Australia	6	1.3.0
CN	China	3	1.3.5
IN	India	3	1.3.0
DE	Germany	3	1.3.4
RU	Russia	3	1.3.4
PL	Poland	3	1.3.0
CA	Canada	3	1.3.5
IL	Israel	2	1.3.4
CZ	Czech Republic	2	1.3.4

6.3 Geographic Insights

* The US dominates both crash (47 events) and hang (27 events) signals, reflecting its large install base and higher participation in Insider Preview builds.
* The 1.3.5/1.3.6 crash surge reached 13 markets simultaneously, indicating a product-level regression rather than an environment-specific issue.
* Brazil, Taiwan, and Germany each saw significant crash spikes in 1.3.6 -- markets largely unaffected in earlier versions.
* CmdPal hangs show earlier geographic spread (from 1.3.0) vs crashes, consistent with the hang being a persistent unresolved defect from launch.
* Smaller markets (DK, RO, IE, ZM, AL) signal broad global distribution of the app even during this turbulent release period.

7 Root Cause Analysis

RCA-1: CmdPalExtension HANG_QUIESCE (Hash: e584f61b)

Failure class: Application hang during quiesce (shutdown/teardown sequence).
Component: CmdPalExtension -- the command palette extension host process.

The HANG_QUIESCE failure occurs when the extension process does not respond to a shutdown signal within the OS timeout window. The single stable hash across all affected versions (1.3.0-1.3.6) confirms this is one code path. Likely causes:

* Deadlock or infinite wait in the CmdPalExtension shutdown path -- possibly waiting on a COM call or async operation that never completes.
* Unresponsive message pump: the extension's UI thread may be blocked on a synchronous operation (disk I/O, COM activation) during teardown.
* Missing CancellationToken propagation in async extension lifecycle methods, causing pending tasks to block shutdown indefinitely.

Resolution evidence: The hang was eliminated in 1.3.7 (zero CmdPal events). The nature of the fix should be documented to prevent re-introduction.

RCA-2: Crash Surge in 1.3.5 / 1.3.6 (Null-Hash Events)

Failure class: Unhandled exception in main app process (PRAID: app).
The majority of crash events carry null (empty) failure hashes, meaning the crashes were not symbolicated or caught by a named exception handler. Null-hash crashes arise from:

* SEH/C++ exception escaping the top-level catch block in WinMain or App::OnLaunched.
* Stack overflow or heap corruption preventing normal exception handling.
* Process termination by the OS (memory pressure, watchdog) without WER bucket generation.
* Missing or expired PDB symbols preventing Partner Center from bucketing the crash.

The surge from 1 event (1.3.4) to 43 events (1.3.5) correlates with a code change or new dependency in 1.3.5. Breadth across 10+ markets and multiple OS versions rules out an environment-specific trigger -- this is a product regression.

RCA-3: WinUI Composition FATAL_USER_CALLBACK_EXCEPTION (Hash: bf902711)

Failure class: Fatal exception in WinUI 3 Composition layer during initialization.
Exception code: 0xc000041d (STATUS_CALLBACK_RETURNED_TRANSACTION).
Module: dcompi.dll -- Microsoft.UI.Composition runtime DLL.

Exception code c000041d fires when a COM callback returns while a kernel transaction (KTM) is active on the thread. This crash is confined to OS build 10.0.26300 (Insider Preview), suggesting a behavioral change in the Composition runtime. The app may be calling Composition APIs from a callback context that was safe on earlier builds but is now prohibited. Root cause is likely a WinUI 3 SDK version mismatch or reentrancy from a COM callback such as DeviceLost or SurfaceContentsLost.

RCA-4: E_DELAYLOAD_MOD_NOT_FOUND / dcompi.dll (Hash: d69abbf3)

Failure class: DLL delay-load failure at startup.
This is a dependency resolution failure, not a code defect in ValleySoft Disk Analyzer. dcompi.dll is a WinUI 3 runtime component that must be present on the device. On OS build 10.0.26220 (OEM/External Preview), the runtime package was absent, corrupted, or at an incompatible version. The app must implement graceful degradation: catch the load failure and surface a user-friendly error message rather than hard-crashing silently.

8 Prioritized Engineering Recommendations

Recommendations are ordered by impact (event volume x user severity x market breadth). Priority: RED (P1) = Critical/ship-blocker, ORANGE (P2) = High/next-sprint, GREEN (P3) = Medium/upcoming sprint.

P1

Resolve Residual Null-Hash Crashes in 1.3.7

Root-cause the 5 unnamed crash events persisting in 1.3.7 (US, Win11 25H2). Enable full WER symbol upload and PDB indexing so future crashes get bucketed. Add a top-level SEH handler in App::OnLaunched and WinMain with structured logging. Integrate Application Insights or a crash SDK to capture stack traces on affected devices.

P1

Document and Regression-Test the CmdPalExtension HANG_QUIESCE Fix

1.3.7 eliminated hash e584f61b. Add an automated test exercising the CmdPalExtension shutdown path under load (simulate command palette open then app close). Ensure coverage across all OS versions where the hang was observed, including Win10 22H2 (10.0.19045). Add shutdown timeout logging to detect future quiesce regressions early.

P2

Fix WinUI Composition Compatibility with Insider Build 10.0.26300

Audit all COM callback handlers calling into Microsoft.UI.Composition APIs -- specifically DeviceLost, FrameCreated, or SurfaceContentsLost handlers. Test against the latest WinUI 3 NuGet (1.6.x+) which may include runtime fixes for build 26300. File an issue with the WinUI team if the crash is in the SDK rather than application code.

P2

Investigate the 1.3.5/1.3.6 Crash Regression Root Cause

The jump from 1 crash event (1.3.4) to 43 (1.3.5) demands root-cause analysis. Diff 1.3.4 vs 1.3.5 for changes to initialization, disk scanning, and exception handling. Enable minidump collection via WER LocalDumps. Wrap all async void event handlers in try/catch -- unhandled exceptions in async void crash the process without generating a named WER bucket.

P2

Implement Graceful Degradation for dcompi.dll Load Failures

Wrap WinUI 3 runtime initialization in a structured exception handler. If E_DELAYLOAD_MOD_NOT_FOUND is caught, display: "Windows UI runtime component missing -- please check for Windows Updates." Add a pre-launch check for minimum WinUI runtime version. This covers preview-build users on builds 10.0.26220 and above.

P3

Expand OS Coverage in CI/CD Test Matrix

CmdPal hangs and WinUI crashes were observed across 7 distinct OS builds. Add Windows Insider Preview (10.0.26300+) and Win10 22H2 (10.0.19045) to the automated test pipeline. Run integration tests on MSIX-installed builds in clean VMs for each OS tier to catch environment-specific failures before release.

P3

Improve Telemetry Signal Quality (Symbolication & Bucketing)

Over 60% of crash events carry null failure hashes. Actions: (a) Upload PDBs to Partner Center symbol server for every release build. (b) Ensure AppxManifest package publisher matches signing cert for correct WER bucketing. (c) Integrate Application Insights SDK with structured exception reporting. (d) Add version-tagged release notes to Partner Center crash dashboard.

P3

Establish Market-Aware Staged Rollout Policy

The US market absorbs 50%+ of crash events, making it a reliable early-signal cohort. Implement staged rollout: release to 5% of US devices first; monitor crash rate for 48h before expanding. Set automated pause thresholds: crash rate > 0.5% or any new failure hash with more than 2 events halts the rollout. Include Brazil, Germany, and India in the first-wave monitoring cohort given their elevated event counts in 1.3.5-1.3.6.

Data sourced from Microsoft Partner Center telemetry (Daily Health.csv, health.csv, Report.csv). All event counts reflect aggregated retail-channel telemetry for x64 PC. This report was generated automatically on July 31, 2026. Stability assessments are based on observed failure counts and do not constitute a formal quality gate.